TopTreeSegR: Mesh-based topological segmentation for plot-wise TLS forest inventory

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Abstract

Terrestrial Laser Scanning (TLS) has revolutionized forest monitoring by capturing millions of three-dimensional points that encode detailed structural information at the individual tree level. However, operational adoption of TLS in forest inventory remains hindered by the lack of automated, scalable, and generalizable segmentation methods. Existing approaches suffer from fundamental limitations: intensive parameter tuning, dataset-specific assumptions, prohibitive computational demands, or reliance on extensive labeled training data.

We present *TopTreeSegR*, a novel mesh-based topological segmentation framework for plot-wise forest TLS point clouds. Departing from point-centric methods, *TopTreeSegR* constructs an α-complex mesh representation that captures the intrinsic topology of the forest scene. Through discrete Morse theory, we compute a discrete gradient vector field on the mesh, identifying critical minima. Density-based seed detection then identifies tree trunks by finding local density peaks among minima near the ground. Tracing ascending manifolds from these seeds through the Morse-Smale complex partitions the canopy into individual tree segments, where each minimum's ascending region defines a complete tree. The framework requires zero parameter tuning across diverse forest types and incorporates plot-wise optimization for operational efficiency.

Through systematic evaluation—α-sensitivity analysis, Bayesian Boundary Refinement assessment, parameter interaction analysis, and validation on the TreeScanPL10k dataset (30 plots, 6 forest types, 1616 manually labeled trees)—*TopTreeSegR* achieves a mean Adjusted Rand Index of 0.695 (±0.118) and F1 score of 0.788 (±0.088), with strong agreement between predicted and ground truth tree counts (R^2 = 0.94). Mean processing time across all plots was 1213 seconds (range: 331–7449 seconds), demonstrating operational viability for routine forest inventory with processing times.

TopTreeSegR is implemented as an open-source R package, providing a plug-and-play solution for individual tree segmentation that works across forest types without parameter tuning. The mesh-based topological approach offers a new paradigm for integrating computational geometry with forest ecology, with direct applications in TLS-based forest monitoring.

Keywords: Terrestrial Laser Scanning · Individual Tree Segmentation · Topological Data Analysis · Discrete Morse Theory · Alpha Complex · Forest Inventory

1. Introduction

Forests cover approximately 31% of Earth's land surface and harbor 80% of terrestrial biodiversity, while sequestering 2.4 ± 0.4 Pg C annually—a critical buffer against anthropogenic carbon emissions (FAO, 2020; Pan et al., 2011). Accurate forest inventory is therefore foundational to climate change mitigation, biodiversity conservation, and sustainable resource management. Traditional field-based inventories, relying on manual measurements of diameter at breast height (DBH), tree height, and species identification, are expensive, time-consuming, and logistically challenging in remote or complex terrain (Liang et al., 2018; Zheng et al., 2016). Moreover, these methods provide only aggregated plot-level estimates, obscuring the fine-scale structural heterogeneity that drives ecosystem function (Calders et al., 2020).

Terrestrial Laser Scanning (TLS) has emerged as a transformative technology that addresses these limitations. By emitting millions of laser pulses per second and measuring the time-of-flight of reflected signals, TLS generates dense three-dimensional point clouds that capture detailed structural information at the individual tree and branch level (Liang et al., 2018). A single TLS scan can acquire >100 million points, encoding tree locations, stem geometry, crown architecture, and even fine branch structure (Hackenberg et al., 2015; Raumonen et al., 2013). This capability enables derivation of forest inventory metrics—tree height, DBH, stem density, biomass—with unprecedented precision and completeness (Burt et al., 2019; Calders et al., 2015).

Despite this transformative potential, the operational adoption of TLS in forest monitoring has been remarkably slow. The fundamental bottleneck is not data acquisition—TLS instruments are increasingly affordable and portable—but rather data processing. The billions of points comprising a typical TLS survey must be segmented into individual tree point clouds before any meaningful inventory metrics can be extracted (Calders et al., 2020; Martin-Ducup et al., 2021). Individual tree segmentation (ITS) from TLS point clouds remains an open challenge, hindered by structural complexity, occlusion and shadowing, overlapping canopies, scalability constraints, and domain specificity that requires laborious parameter tuning for each new forest type.

Existing ITS methods fall into several broad categories, each with fundamental limitations. Bottom-up segmentation (Burt et al., 2019; Trochta et al., 2017) identifies tree stems at low heights and grows crowns upward, but requires careful parameter tuning that varies systematically with forest type, stem density, and point quality. Top-down approaches (Wang et al., 2020) use hierarchical deep learning frameworks with semantic labeling and graph-based optimization for tree instance recognition, but are typically designed for Unmanned Aerial Vehicle-LS data in urban environments, making their application to understory tree detection and crown delineation in dense, multi-layered TLS forest scenes challenging. Deep learning methods (Chang et al., 2022; Krisanski et al., 2021; Xi and Hopkinson, 2021) have shown remarkable performance but face prohibitive barriers: the need for large labeled training datasets for each forest type, computational costs, domain shift issues, and limited interpretability. Allometric methods (Burt et al., 2019) incorporate ecological knowledge but are species- and site-specific, limiting generality. Topological methods (Xu et al., 2023) represent a promising direction, leveraging discrete Morse theory to identify meaningful structures, but prior implementations have focused on methodological demonstration and robustness evaluation rather than operational deployment.

In this work, we introduce a fundamentally different approach: mesh-based topological analysis. Rather than operating directly on discrete points, we construct a continuous mesh representation—the α-complex (Edelsbrunner, 2014)—a subset of the Delaunay triangulation (Delaunay, 1934) that interpolates the point cloud and encodes the connectivity relationships of the forest scene. Through discrete Morse theory (Forman, 1998), we compute a discrete gradient vector field on the mesh, identifying critical minima. Density-based seed detection then identifies tree trunks by finding local density peaks among minima near the ground. Tracing ascending manifolds from these seeds through the Morse-Smale complex (Edelsbrunner et al., 2003; Edelsbrunner and Harer, 2010) partitions the canopy into individual tree segments, where each minimum's ascending region defines a complete tree. This topological approach offers automatic operation with zero parameter tuning, natural handling of overlapping canopies, and scalability through mesh simplification and parallel processing.

We present *TopTreeSegR*, an open-source R package implementing this mesh-based topological segmentation framework for plot-wise forest TLS point clouds. The framework integrates three core computational components: (1) *ahull3D* for α-complex mesh construction, (2) *DiscreteMorseR* for discrete Morse theory analysis and Forman gradient computation, and (3) the *TopTreeSegR* pipeline with density-based seed detection, ascending manifold tracing, and validation through label propagation. Our contributions are fivefold:

(i) we demonstrate that mesh-based topological analysis provides a robust, parameter-free foundation for individual tree segmentation that generalizes across diverse forest types;
(ii) we show that structural information loss during meshing is negligible for key forest inventory metrics—height, DBH, and crown volume—confirming operational applicability;
(iii) we conduct comprehensive parameter analysis including α-sensitivity, Bayesian Boundary Refinement assessment, and parameter interaction analysis to establish optimal configuration;
(iv) we validate the framework on the TreeScanPL10k dataset comprising 30 plots from six Polish forests representing contrasting ecological conditions (Stereńczak et al., 2026); and
(v) we provide an open-source, production-ready R package that makes high-precision TLS-based forest inventory accessible to the broader ecological community.

2. Theoretical background

TopTreeSegR is built on three key concepts from computational topology that work together to segment individual trees from TLS point clouds: the α-complex for mesh construction, discrete Morse theory for topological analysis, and Bayesian inference for boundary refinement.

2.1. From point cloud to mesh: the α-complex

The TLS point cloud is a discrete set of points with no inherent connectivity. To analyze the structure of the forest—how stems connect to branches, how crowns overlap—we first construct a continuous mesh representation called α-complex. The α-complex (Edelsbrunner, 2014) is a subset of the Delaunay triangulation that captures the shape of the point cloud at different levels of detail (Delaunay, 1934). A single parameter, α, controls the scale: small α captures fine branch structure, while larger α connects tree components and fills gaps in crown data.

In *TopTreeSegR*, the α-complex serves two critical purposes. First, it provides a mesh that encodes the topological connectivity of stems, branches, and crowns—information that is not directly available in the raw point cloud. Second, it dramatically reduces computational complexity: a point cloud of 1.2 million points may be represented by a mesh with only 134,000 vertices—a ~90% reduction in data volume. This mesh simplification is essential for fast processing of large TLS plots while preserving the tree structure.

2.2. Analyzing structure: discrete Morse theory and the Morse-Smale complex

Once the mesh is constructed, we need to identify which parts of the mesh belong to which tree. For this, we use discrete Morse theory (Forman, 1998), a mathematical framework that analyzes the topology of a mesh using a scalar function defined on its vertices. We define the height function $f(p) = p_z$—the elevation of each vertex. The theory computes a Forman gradient—a discrete vector field that points in the direction of increasing height. The critical simplices of this gradient characterize the topology:

- 0-simplices (minima): vertices not paired with edges, representing local minima of the height function.
- 1-simplices (saddles): edges not paired with vertices or triangles, representing saddle points.
- 2-simplices (maxima): triangles not paired with edges, representing local maxima.

From the Forman gradient, we construct the Morse-Smale complex (Edelsbrunner et al., 2003; Edelsbrunner and Harer, 2010)—a topological structure that partitions the mesh into regions of uniform gradient flow. The complex is defined by the paths connecting critical simplices: minima to saddles, saddles to saddles, and saddles to maxima. The ascending manifold of a minimum is the set of vertices that flow to that minimum along the gradient.

In forest segmentation, the ascending manifolds of minima define the influence zones of individual trees. Each point in the canopy belongs to the ascending manifold of exactly one minimum, providing a natural, mathematically well-founded partitioning of the forest scene. The Morse-Smale complex thus serves as the foundation for region growing: starting from seed minima (tree trunks), we trace their ascending manifolds upward to capture complete tree crowns.

However, not all minima correspond to distinct trees—within a single tree, branch tips and low-hanging branches also produce minima. To address this, *TopTreeSegR* first partitions the mesh into connected components using *DiscreteMorseR::get_CCMESH()*. Each component represents a spatially isolated group of mesh vertices—a tree cluster that may contain one or multiple trees. Components that contain at least one minimum below the stem height threshold (0.5m) are retained as potential tree clusters; components without such minima are discarded.

Within each retained component, density-based seed detection identifies tree trunks by finding local density peaks among minima near the ground. This approach automatically adapts to variable tree spacing and eliminates the need for manual thresholding or dataset-specific calibration. The ascending manifolds of these selected seed minima—traced through the Morse-Smale complex—then define the complete individual trees, effectively separating trees within the same connected component. Finally, all segmented trees from all components are combined into a single labeled mesh (i.e., point cloud) for the entire plot.

2.3. Refining boundaries: Bayesian inference

After initial segmentation, boundary points between adjacent trees—particularly in dense forests with overlapping crowns—may be ambiguous. *TopTreeSegR* applies Bayesian Boundary Refinement (BBR) to resolve these ambiguities probabilistically.

For each boundary point, the posterior probability of belonging to each neighboring tree is computed as:

$$P(\text{label} = i \mid \text{data}) \propto P(\text{data} \mid \text{label} = i) \cdot P(\text{label} = i)$$

where:

- Likelihood $P(\text{data} \mid \text{label} = i)$: The probability of observing the point's elevation given it belongs to tree i. This is modeled as a Gaussian distribution centered on the mean elevation of tree i, capturing the

expectation that points within a tree have coherent height distributions. The *likelihood_strength* parameter (default: 1.6) controls the influence of this term.

- Prior $P(\text{label} = i)$: Captures spatial coherence through neighborhood relationships applying a Markovian prior, (Geman and Geman, 1984), encoding the assumption that neighboring points tend to belong to the same tree. The *prior_strength* parameter (default: 1.0) controls the strength of this spatial smoothness.

The point is assigned to the tree that maximizes the posterior probability (Maximum A Posteriori or MAP estimation), ensuring elevation-consistent and spatially coherent boundaries (Gelman et al., 2013). The BBR is applied in two passes—aggressive then conservative—to achieve optimal accuracy.

3. Methodology

TopTreeSegR implements a fully automated pipeline for individual tree segmentation from plot-wise TLS point clouds. The framework integrates three core R packages: *ahull3D* for mesh construction, *DiscreteMorseR* for topological analysis, and the *TopTreeSegR* pipeline for seed detection, segmentation, and refinement (Fig. 1).

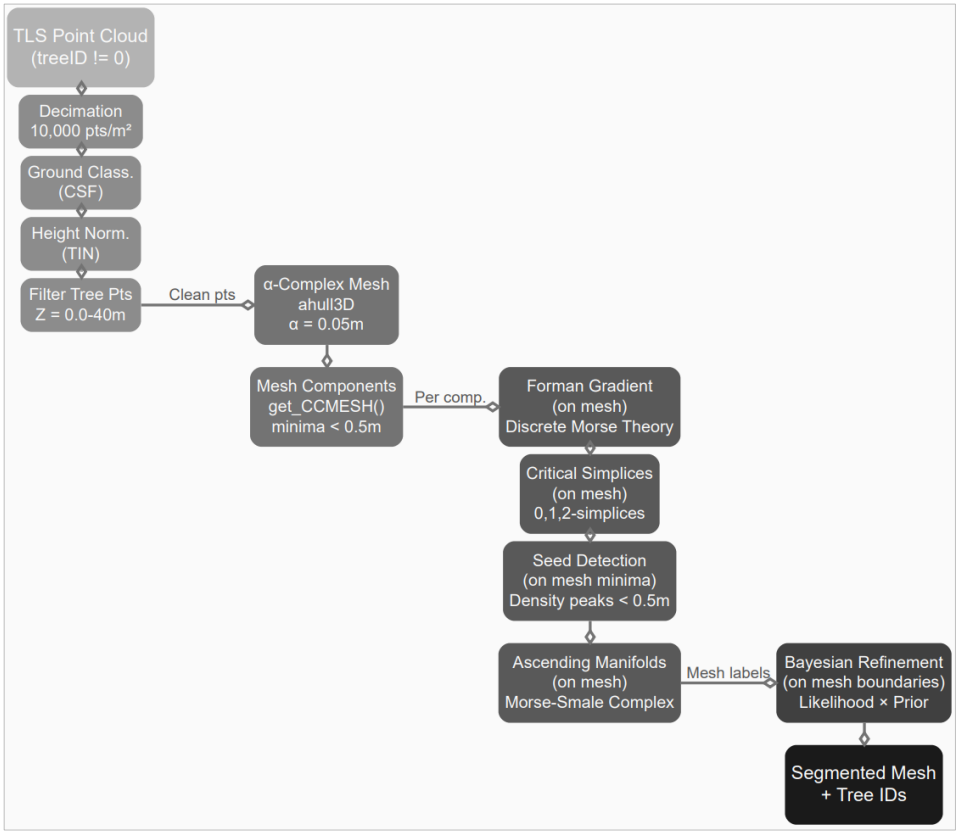


Figure 1: *TopTreeSegR* processing pipeline for individual tree segmentation from TLS point clouds. The workflow comprises four stages: (1) preprocessing—decimation, ground classification (CSF), height normalization (TIN), and tree point filtering (Z = 0.0–40 m); (2) mesh construction—α-complex generation (*ahull3D*) and connected component extraction (*DiscreteMorseR::get_CCMESH()*); (3) topological segmentation—Forman gradient computation and critical simplex (0-, 1-, 2-simplex) identification (*DiscreteMorseR*), density-based seed detection from minima below 0.5 m, and ascending manifold tracing through the Morse-Smale complex (*TopTreeSegR*); and (4) refinement and output—Bayesian Boundary Refinement (likelihood × prior) and final tree-labeled mesh (*TopTreeSegR*). The framework operates automatically across diverse forest types without parameter tuning.

3.1. Preprocessing

The input TLS point cloud—containing only tree points (treeID != 0)—undergoes four preprocessing steps using the *lidR* package (Roussel et al., 2020): (1) spatial decimation to 10,000 points/m² via homogenization, (2) ground classification using the Cloth Simulation Filter (Zhang et al., 2016), (3) height normalization to above-ground elevation using TIN interpolation (Peucker et al., 1978) from ground points, and (4) filtering to retain points within Z = 0.0–40 m. This preprocessing reduces point count by ~90–95% while preserving essential tree structure.

3.2. Mesh construction

The preprocessed point cloud is converted to an α-complex mesh using *ahull3D::ahull3d()* (α = 0.05 m). The mesh is then partitioned into connected components using *DiscreteMorseR::get_CCMESH()*. Components containing at least one minimum below the stem height threshold (0.5 m) are retained as potential tree clusters; components without such minima are discarded. All retained components are processed in parallel (*select_largest = FALSE*), ensuring no trees are missed.

3.3. Topological segmentation

For each retained component, topological analysis and segmentation are performed on the mesh as described in Section 2.2. Briefly, the Forman gradient of the height function is computed and critical simplices are identified. Density-based seed detection then locates tree trunks, and ascending manifolds traced through the Morse-Smale complex partition the component into individual trees. Segmented trees from all components are combined into a single labeled point cloud.

3.4. Bayesian Boundary Refinement

Initial tree boundaries are refined on the mesh using the BBR algorithm (Section 2.3), applied in two passes—aggressive then conservative—to achieve optimal accuracy. The final output is a labeled mesh with tree IDs assigned to each vertex.

3.5. Validation framework

TopTreeSegR provides a comprehensive validation suite via *TopTreeSegR::validate_TTS()*. Given ground truth labels (treeID) and predicted labels, the framework computes:

- Adjusted Rand Index (ARI) (Rand, 1971): measures agreement between predicted and ground truth clusterings, accounting for chance;
- F1 score (Hand et al., 2021; Rijsbergen, 1979): harmonic mean of precision and recall.

For the α-sensitivity analysis (Section 5.1), tree-level structural metrics were derived from the segmented mesh vertices to assess structural fidelity across mesh resolutions. DBH was estimated by fitting a circle to breast-height points (Z = 1.15–1.45 m) using a least-squares circle fitting algorithm. Points within this height interval were projected to the XY-plane, and the circle center and radius were estimated by solving a linear system of equations (Taubin, 1991). DBH was then calculated as twice the radius and converted to centimeters. Crown volume was estimated using a voxel-based approach with a voxel size of 0.3 m (Calders et al., 2015). The point cloud was partitioned into a regular 3D grid, and crown volume was calculated as the sum of occupied voxel volumes for points above 60% of the tree height.

The validation framework automatically aligns predicted labels to ground truth labels using the Hungarian algorithm (Kuhn, 1955). This resolves the label correspondence problem, ensuring that predicted tree IDs are correctly matched to ground truth tree IDs before computing evaluation metrics.

4. Experimental design

We evaluate *TopTreeSegR* through a systematic multi-stage experimental design using the Wytham Woods reference dataset (three configurations) and the TreeScanPL10k dataset (30 plots from six Polish forests).

4.1. Wytham Woods reference dataset

Our Wytham Woods reference dataset (Calders et al., 2022) comprises 12 selected leaf-off trees with three TLS point cloud configurations from a temperate deciduous forest in Oxford, UK (51° 46′ N, 1° 20′ W) (Fig. 2). The dataset was acquired using a Leica ScanStation P30 TLS instrument with a pulse rate of 1,000,000 points/second and angular resolution of 0.036°. The three configurations serve distinct experimental purposes. Configuration 1 containing 12 individual trees with varying degrees of crown overlap (Fig. 2a–b). Configuration 2 (Fig. 2c–d)—a subset of configuration 1—has 6 well-separated trees with trunk distances ≥1 m, exhibiting minimal to medium crown overlap. Configuration 3 (Fig. 2e–f)—the other subset of configuration 1—shows 6 adjacent trees with touching trunks and overlapping crowns, representing the most challenging scenario for individual tree segmentation.

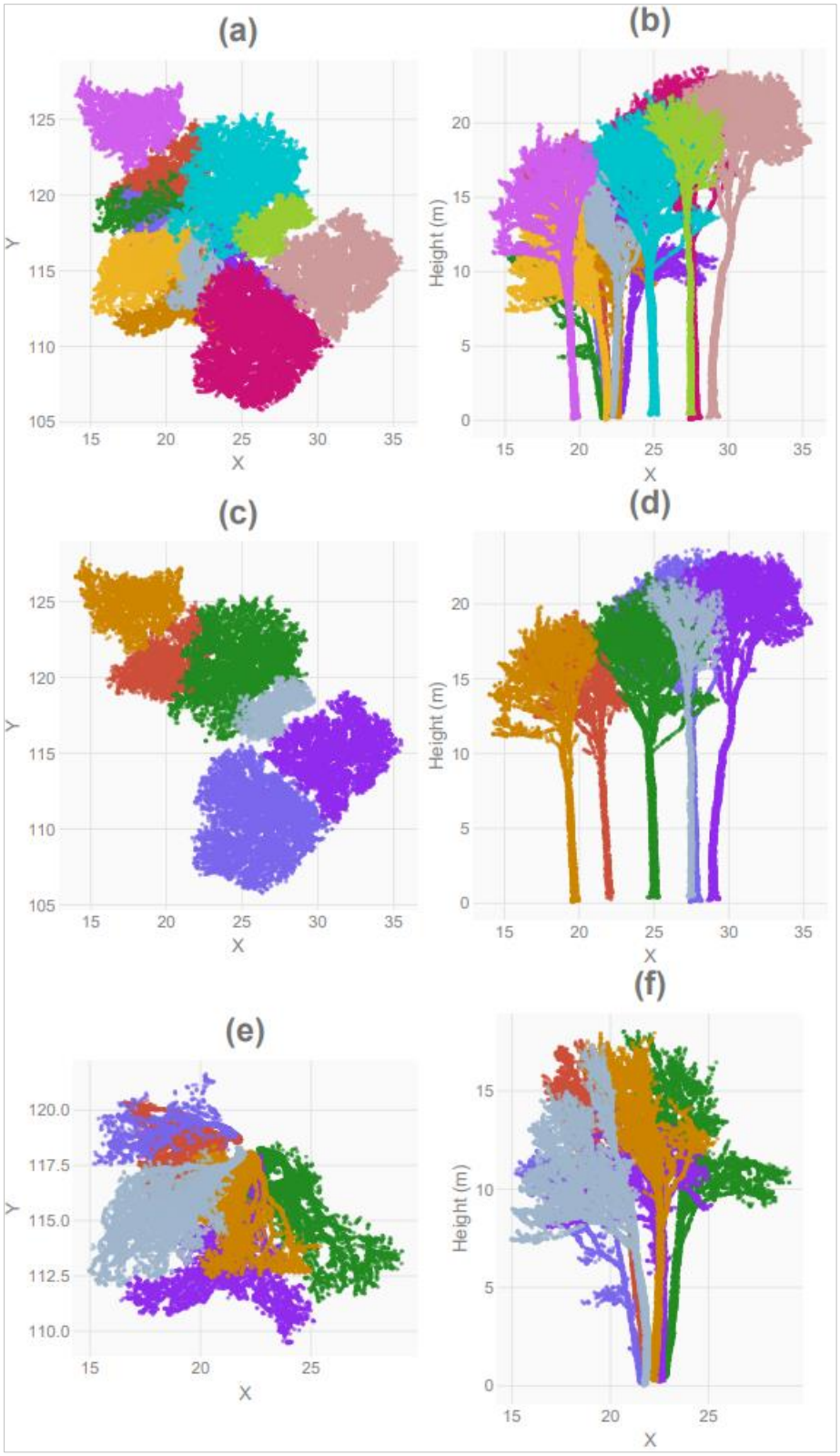


Figure 2: Terrestrial LiDAR point clouds showing three tree configurations from Wytham Woods, UK. Panels (a)–(b): 12 trees showing XY (left) and XZ (right) projections. Panels (c)–(d): 6 well-separated trees (≥1m trunk distance). Panels (e)–(f): 6 adjacent trees (touching trunks). Points decimated to ~1,000 pts/m^2 for visualization. Each tree is color-coded individually.

All configurations were manually labeled by trained foresters, providing ground truth tree IDs for each point. The reference dataset contains 12 labeled trees across the three configurations.

4.2. TreeScanPL10k dataset

The TreeScanPL10k dataset (Stereńczak et al., 2026) comprises 272 circular TLS plots (15 m radius = ~225 m^2) across Polish forests. We selected 30 plots from six forest districts representing diverse ecological conditions across the country (Table 1). Each plot was scanned using a Leica ScanStation P30 with protocols following the EuroSDR guidelines (Liang et al., 2018).

Table 1. Summary of the 30 selected TreeScanPL10k plots by forest district.

District	Plots	Forest type	Dominant species	Elevation (m asl)	Terrain
Gorlice	15	Mixed coniferous-deciduous	Norway spruce (*Picea abies*), European beech (*Fagus sylvatica*)	300–500	Moderate (5–15°)
Herby	7	Pine-dominated	Scots pine (*Pinus sylvestris*), hornbeam (*Carpinus betulus*), oak (*Quercus* spp.)	250–300	Gentle (<5°)
Katrynka	3	Deciduous	English oak (*Quercus robur*), small-leaved lime (*Tilia cordata*), Norway maple (*Acer platanoides*)	150–200	Flat
Milicz	1	Mixed	Scots pine, English oak, silver birch (*Betula pendula*)	100–120	Flat
Piensk	3	Coniferous	Norway spruce, European larch (*Larix decidua*)	400–600	Moderate (5–10°)
Suprasl	1	Mixed	Scots pine, English oak, silver birch	130–150	Flat

Each plot was manually labeled by trained foresters, assigning unique tree IDs (*treeID*) to all visible trees. Species identification was performed for 1,132 trees (70%) representing 27 species, with the most common being Scots pine, Norway spruce, Silver fir, and European beech. The selected dataset contains 1,616 manually labeled trees across the 30 plots, with stem densities ranging from 589 to 1,791 stems/ha and point densities ranging from 0.6 to 5.3 cm (vegetation point density measured as average nearest-neighbor distance). Point counts per plot average 73 million points (range: 9–175 million points). Using the TreeScanPL10k dataset, we evaluate *TopTreeSegR* on all 30 plots with fixed parameters (α = 0.05 m, *stem_height* = 0.5 m, *density_cell* = 0.5 m), as established by the parameter interaction analysis (Section 5.3). Segmentation accuracy (ARI and F1) is computed per plot to assess performance across the full range of forest types and structural complexities represented in the dataset. For this validation, we used the default *TopTreeSegR::TTS_pipeline()* (the main function, which includes the first BBR pass by default), as the optional second pass provides only modest additional improvement (Section 5.2).

4.3. Parameter interaction and sensitivity analyses

To establish optimal parameter configurations and understand parameter dependencies, we conducted three complementary experiments using the Wytham Woods configurations:

- α-Sensitivity Analysis (Fig. 2a–b): Using the 12-tree configuration, we evaluated segmentation accuracy (ARI, F1) across α values of 0.02, 0.05, 0.08, 0.10, 0.12, 0.15, and 0.20 m to identify the optimal α value for mesh construction and to assess forest plot (i.e., tree) structure representability by the given mesh.
- BBR Assessment (Fig. 2c–d): Using the 6 well-separated trees configuration, we quantified the improvement from BBR by comparing segmentation accuracy from *TTS_pipeline()* against *TTS_pipeline() + BBR* (applying the second conservative pass). The first pass (*confidence_threshold* = 1.0) aggressively corrects major boundary errors, while the second pass (*confidence_threshold* = 1.9)

refines only highly probable improvements (Table 2). This comparison quantifies the additional accuracy gain from the second refinement pass, as well as the robustness of *TTS_pipeline()* on normally distributed trees within the plot.

- Parameter Interaction Analysis (Table 2; Fig. 2e–f): Using the 6 adjacent trees configuration—the most challenging scenario—we conducted a full-factorial experiment examining density-based stem detection robustness (i.e., *TTS_pipeline()* robustness) and dependencies among *alpha*, *stem_height*, and *density_cell*. The 3 × 4 × 4 design (48 parameter combinations) evaluated *alpha* at 0.03, 0.05, and 0.07 m, *stem_height* at 0.2, 0.3, 0.5, and 0.7 m, and *density_cell* at 0.3, 0.5, 0.7, and 1.0 m. Bayesian refinement parameters (*prior_strength*, *likelihood_strength*, *confidence_threshold*) were held at default values during this analysis.

Table 2. *TopTreeSegR::TTS_pipeline()* parameters, their default values with corresponding implications and ranges.

Name	Default value	Implication	Range or values
*las**	-	Input LAS point cloud (forest plot)	[-]
*method**	morse-smale	Segmentation method	morse-smale, gradient
alpha	0.1	Alpha parameter for α-complex mesh construction; controls level of detail	0.03–0.20 (m)
*input_truth**	pid	Point ID attribute from LAS file	[-]
stem_height	0.5	Height threshold for detecting stem seeds (minima below this height considered potential trunks)	≥ 0.3 (m)
density_cell	1.0	Cell size for density-based seed detection; controls sensitivity of trunk detection	≥ 0.5 (m)
prior_strength	1.0	Spatial consistency strength in Bayesian refinement; controls how much points match neighbors' labels	0.5–2.0
likelihood_strength	1.6	Elevation consistency strength; exponent controlling how strongly points at similar heights share labels	0.5–2.5
confidence_threshold	1.0	Aggressiveness of boundary refinement; higher values = more conservative changes	1.0–2.0
cores	2	Number of CPU threads for parallel processing	≥ 2

* String input parameters.

4.4. Computational setup

All experiments were conducted on a workstation with dual Intel Xeon Gold 6154 CPUs (16 cores, 32 threads total) at 3.70 GHz base speed, 128 GB RAM, running Windows 11. Parallelization was implemented using the *clustermq* package (Schubert, 2019) across 28 cores, as specified in the *TopTreeSegR* pipeline (cores = 28). Processing time (preprocessing + segmentation + refinement) was recorded for each plot.

5. Results

5.1. α-sensitivity analysis

Using the 12-tree Wytham Woods configuration, we evaluated the effect of α on segmentation accuracy and mesh complexity across values from 0.02 to 0.20 m (Fig. 3a–b). Increasing α from 0.03 to 0.20 m reduced mesh vertices by 78% (50,473 to 11,158) and processing time from 224 to 84 seconds. Height RMSE remained negligible (<0.001 m) across all α values. DBH RMSE increased from 0.61 cm at α = 0.03 m to 1.66 cm at α = 0.20 m, while crown volume RMSE increased from 0.33 m^3 to 35.34 m^3. The optimal balance between computational efficiency and structural fidelity was achieved at α = 0.05–0.07 m, with both selected for subsequent experiments and with α = 0.05 m selected for the plot-wise robustness validation.

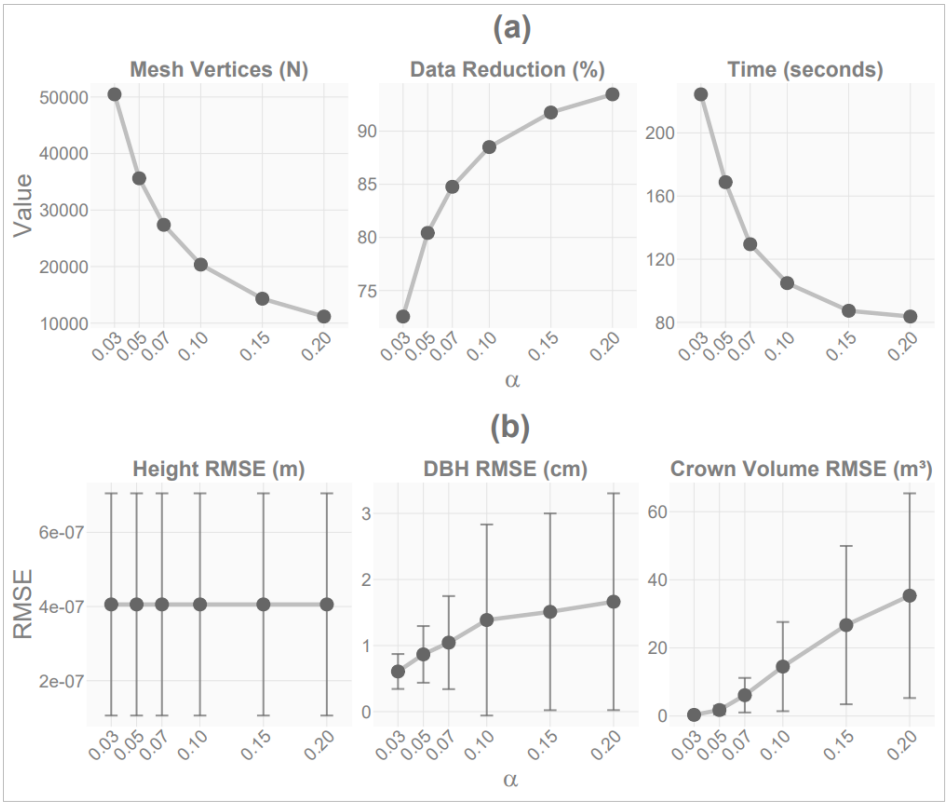


Figure 3: α-sensitivity analysis on the 12-tree Wytham Woods configuration. (a) Number of mesh vertices (N), corresponding data reduction (%) and processing time (s) across α values from 0.02 to 0.20 m. (b) Height RMSE (m), DBH RMSE (cm), and crown volume RMSE (m^3) across α values. Error bars indicate standard deviation across 12 trees.

5.2. Bayesian Boundary Refinement assessment

Using the 6 well-separated trees configuration, we evaluated the improvement from BBR by comparing segmentation accuracy from *TopTreeSegR::TTS_pipeline()* against *TTS_pipeline()* + *BBR* (Fig. 4a–b). At α = 0.05 m, BBR improved ARI modestly from 0.825 to 0.864 (+0.039) at the cost of 66 additional seconds (247 to 313 seconds). At α = 0.07 m, BBR improved ARI from 0.829 to 0.863 (+0.034) with 34 additional seconds (164 to 198 seconds). The consistent but small improvement across both α values indicates that while the second conservative BBR pass provides some refinement to boundary assignments, the primary benefit is already captured by the default first pass.

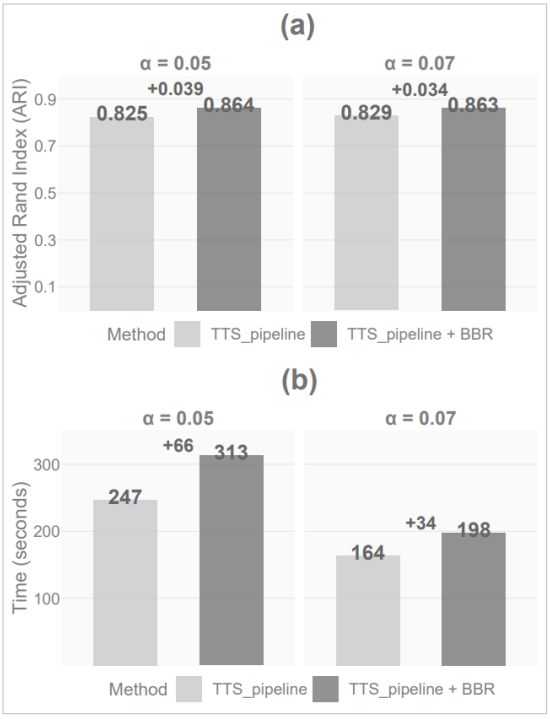


Figure 4: BBR assessment analysis results using the 6 well-seperated Wytham Woods trees configuration. (a) ARI at $\alpha = 0.05$ and 0.07 m for *TTS_pipeline()* vs. *TTS_piplene()* + *BRR*. (b) Processing time (s) at α = 0.05 and 0.07 m for *TTS_pipeline()* vs. *TTS_piplene()* + *BRR*.

5.3. Parameter interaction analysis

Using the 6 adjacent trees configuration with touching trunks and overlapping crowns (Fig. 5a–b), we examined dependencies among *alpha*, *stem_height*, and *density_cell*. *density_cell* ≥0.7 m consistently produced zero ARI (no trees detected), indicating the density cell size was too coarse to capture individual tree trunks. For successful combinations (*density_cell* = 0.3 or 0.5 m), *alpha* = 0.05–0.07 m with *stem_height* = 0.3–0.5 m yielded the highest ARI values (0.219–0.226). Processing time ranged from 40 to 162 seconds, increasing with finer mesh and density resolutions.

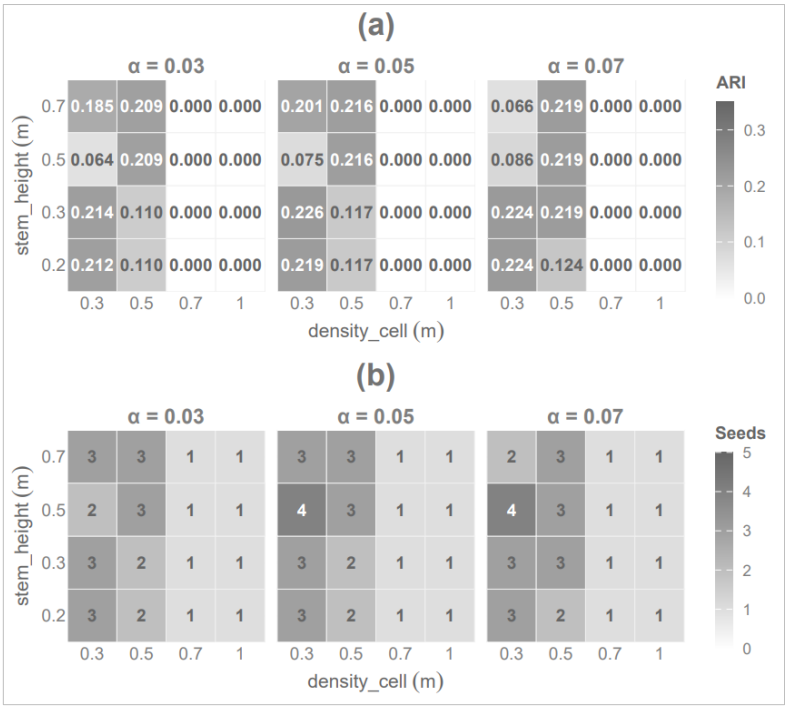


Figure 5: Parameter interaction analysis results using the 6 adjacent Wytham Woods trees configuration. (a) ARI at α = 0.03, 0.05, and 0.07 m across parameter combinations of *density_cell* (m) on the x-axis and *stem_height* (m) on the y-axis. (b) Number of detected seeds at α = 0.03, 0.05, and 0.07 m across parameter combinations of *density_cell* (m) on the x-axis and *stem_height* (m) on the y-axis.

5.4. Plot-wise robustness validation

5.4.1. Aggregate performance

Across the 30 TreeScanPL10k plots, *TopTreeSegR* achieved a mean Adjusted Rand Index of 0.695 (±0.118), ranging from 0.409 to 0.908 (Fig. 6a). The F1 score averaged 0.788 (±0.094), with precision and recall both at 0.788 (not shown). Total processing time averaged 1,213 seconds per plot (range: 331–7,449 seconds), with data reduction from preprocessing averaging 95.1%.

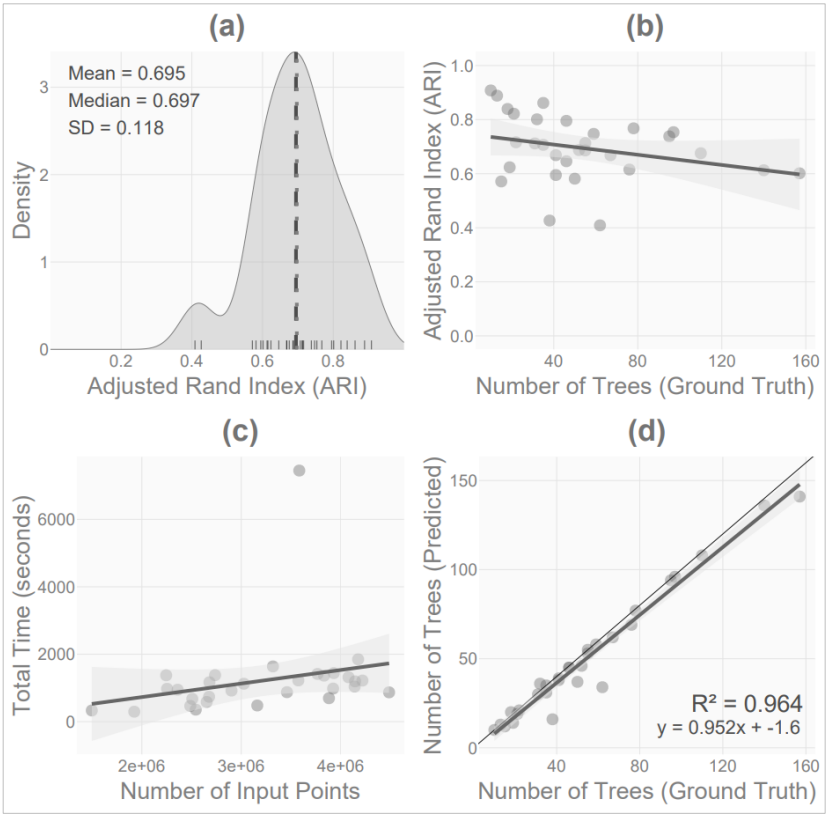


Figure 6: Plot-wise validation of *TopTreeSegR* on 30 TreeScanPL10k plots. (a) Distribution ARI scores across all plots. (b) ARI vs. number of trees (ground truth) with linear regression fit. (c) Processing time vs. number of input points with linear regression fit. (d) Ground truth vs. predicted tree counts with 1:1 reference line and linear regression fit.

ARI showed no strong relationship with the number of ground truth trees (Fig. 6b), with slightly decreasing performance across plots containing 10 to 157 trees, suggesting robustness to plot complexity. Processing time also scaled slightly with the number of input points (Fig. 6c), though with considerable variability (caused by one outlier). Ground truth vs. predicted tree counts showed strong agreement (R^2 = 0.964, y = 0.952x – 1.6; Fig. 6d), indicating that *TopTreeSegR* accurately detected tree numbers (i.e., seeds) despite challenges in boundary assignment.

5.4.2. Per-plot performance

Performance varied considerably across the 30 plots (Fig. 7). The highest ARI was achieved on Rem_Gorlice_2015_5805203.laz (ARI = 0.908, F1 = 0.951), a plot containing only 10 well-separated trees. Conversely, the lowest performance was observed on Rem_Gorlice_2015_7404103.laz (ARI = 0.409, F1 = 0.579), a complex mixed-species plot with 62 trees. Plots from Herby and Katrynka generally showed higher performance (ARI > 0.7), while Piensk plots with high stem densities (140–157 trees) exhibited lower ARI values (0.601–0.613), reflecting the challenges of dense coniferous stands. Tree count agreement was strongest in plots with well-separated trees (e.g., Rem_Gorlice_2015_2603203.laz: 13 vs. 13; Rem_Gorlice_2015_5805203.laz: 10 vs. 10), while significant discrepancies occurred in complex plots where stem detection failed (e.g., Rem_Gorlice_2015_7304103.laz: 38 vs. 16).



Figure 7: Per-plot segmentation performance for all 30 TreeScanPL10k plots, ordered by alphabetical plot name. Each panel displays ARI (black dot) and F1 (grey triangle) on the left, with ground truth (dark grey bar) and predicted (light grey bar) tree counts normalized on the right.

5.4.3. Six representative plots

To illustrate segmentation quality and mesh representation across the performance spectrum, we selected six representative plots: two best-performing (ARI = 0.908, 0.889), two mean-performing (ARI = 0.716, 0.686), and two worst-performing (ARI = 0.427, 0.409) plots (Fig. 8). Each plot is shown as decimated input point cloud (PC) and segmented mesh (MESH) in XY and XZ projections, with individual trees color-coded.

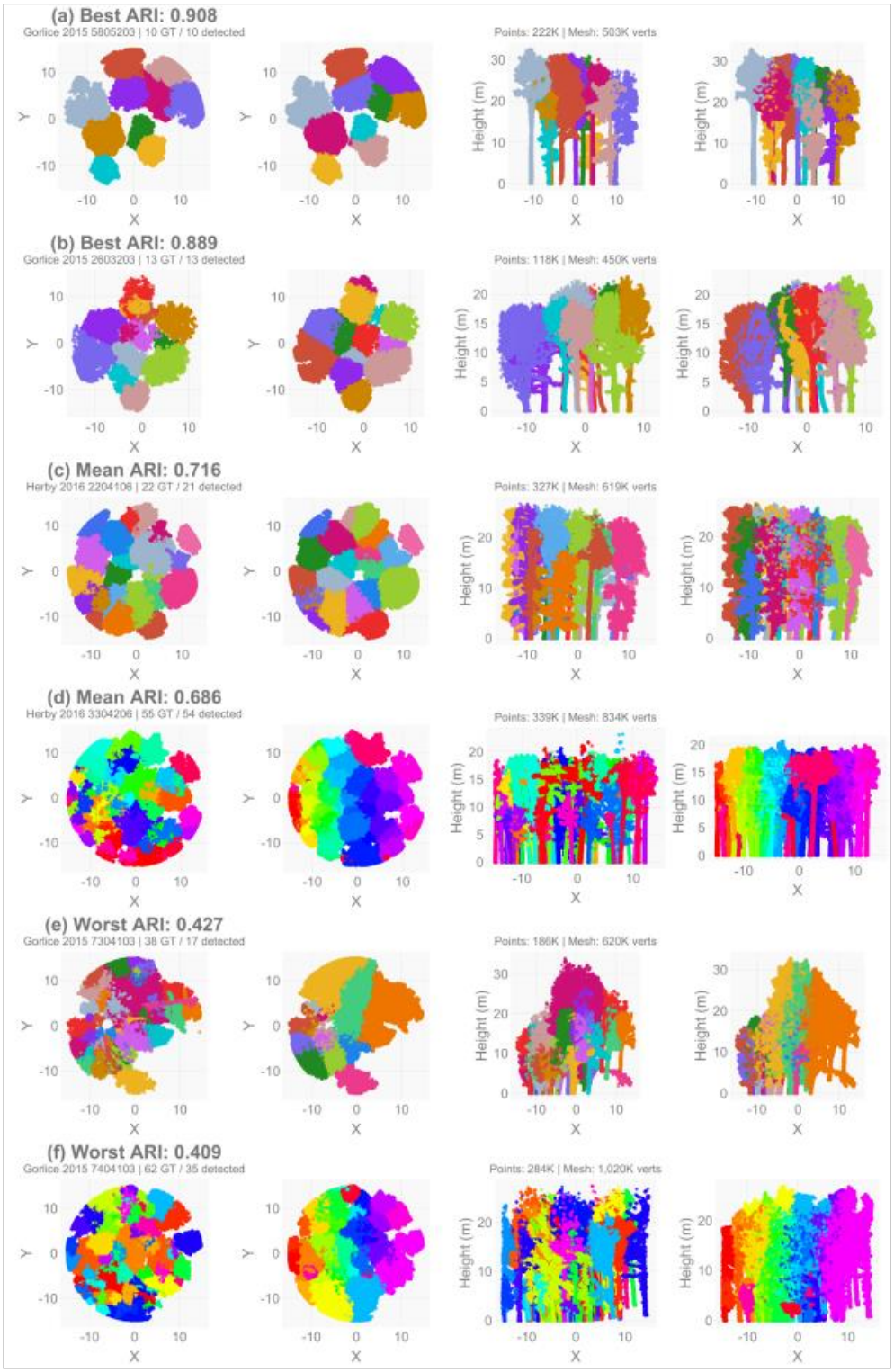


Figure 8: Comparison of decimated input point clouds and segmented meshes for six representative forest plots from TreeScanPL10k. Rows show plots with best, mean, and worst segmentation accuracy (Adjusted Rand Index, ARI): (a) ARI = 0.908, (b) ARI = 0.889, (c) ARI = 0.716, (d) ARI = 0.686, (e) ARI = 0.427, (f) ARI = 0.409. XY (top-down) and XZ (side) projections are shown; point clouds decimated to ~800 pts/m² for visualization. Tree colors indicate individual tree IDs.

The best-performing plots (Rem_Gorlice_2015_5805203.laz: 10 trees; Rem_Gorlice_2015_2603203.laz: 13 trees) exhibited clearly separated tree crowns and well-defined trunks, with the mesh accurately capturing the topological structure of each tree. The mean-performing plots (Rem_Herby_2016_2204106.laz: 22 trees; Rem_Herby_2016_3304206.laz: 55 trees) showed moderate crown overlap, with the mesh generally preserving tree boundaries though some merging occurred in dense crown regions. The worst-performing plots

(Rem_Gorlice_2015_7304103.laz: 38 trees; Rem_Gorlice_2015_7404103.laz: 62 trees) featured dense, multi-layered canopies with overlapping crowns and closely spaced trunks. Here, the density-based seed detector struggled to identify individual stems, resulting in substantial under-segmentation (38 vs. 16, 62 vs. 34 predicted trees) and mesh fragmentation.

These visual comparisons demonstrate that *TopTreeSegR*'s performance is primarily constrained by the ability to detect tree trunks in dense, overlapping stands, while the mesh representation reliably captures tree structure when trunks are successfully identified. This limitation will be addressed in future work through improved seed detection strategies.

6. Discussion

We have presented *TopTreeSegR*, a mesh-based topological segmentation framework for individual tree extraction from plot-wise TLS point clouds. Through systematic evaluation across controlled Wytham Woods configurations and 30 diverse TreeScanPL10k forest plots, we demonstrated that the α-complex mesh representation, combined with discrete Morse theory and density-based seed detection, provides a robust foundation for automated tree segmentation. Our results highlight several key findings with implications for operational forest inventory, as well as important limitations that point toward future research directions.

6.1. Mesh-based topological segmentation: advantages and trade-offs

The mesh-based approach offers fundamental advantages over point-centric segmentation methods. By constructing an α-complex that interpolates the point cloud, *TopTreeSegR* recovers the underlying topology of the forest scene—the connectivity relationships between stems, branches, and crowns that are not directly observable in the discrete point cloud. This topological representation enables the identification of critical features (minima, saddles, maxima) through discrete Morse theory, providing a mathematically well-founded basis for partitioning the canopy into individual trees (Xu et al., 2023).

Our α-sensitivity analysis (Section 5.1; Fig. 3) revealed that α = 0.05 m achieves an optimal balance between computational efficiency and structural fidelity. At this resolution, the mesh preserves fine-scale tree structure while dramatically reducing computational complexity. Critically, from a forest management perspective, structural information loss at α = 0.05 m is negligible: height RMSE remained <0.001 m, DBH RMSE was 0.87 cm, and crown volume RMSE was 1.75 m^3 (Section 5.1; Fig. 3b). This means that the mesh representation at α = 0.05 m captures the essential structural metrics required for operational forest inventory—tree height, diameter, and crown dimensions—with minimal loss of accuracy, while reducing mesh vertices by 80% and processing time by 25% compared to the finest resolution (α = 0.03 m). This demonstrates a key advantage of the mesh-based approach: the α parameter provides a principled way to control the trade-off between accuracy and computational cost without compromising the structural information most critical to forest management applications.

6.2. The role of Bayesian Boundary Refinement

The BBR consistently improved segmentation accuracy beyond the default first pass (Section 5.2; Fig. 4). Applying the optional second conservative pass (*confidence_threshold* = 1.9) yielded +0.039 and +0.034 ARI improvements at α = 0.05 and 0.07 m, respectively. This demonstrates that while the default first pass captures most of the BBR benefit, the second pass provides a modest but consistent refinement, particularly in regions of crown overlap. The additional computational overhead is modest (39–66 seconds per plot, approximately 20% of total processing time), suggesting the two-pass strategy is worthwhile for plots where boundary ambiguity is high. The BBR's reliance on elevation consistency and spatial coherence is particularly effective in forests with clear vertical stratification, where tree height distributions are well-separated.

6.3. The critical role of seed detection in dense forests

The parameter interaction analysis (Section 5.3; Fig. 5) revealed a critical limitation of the current approach: the density-based seed detector fails when trunks are too close together. For the 6 adjacent trees configuration with touching trunks, *density_cell* ≥0.7 m consistently produced zero ARI (no trees detected), while even the best-performing parameter combinations (*density_cell* = 0.3 or 0.5 m, *stem_height* = 0.3–0.5 m) yielded ARI values of only 0.219–0.226—substantially lower than the 0.8+ ARI achieved on well-separated trees.

This performance gap is directly reflected in the plot-wise validation (Section 5.4). The best-performing plots (ARI > 0.85) contained 10–35 well-separated trees with clear trunk structures, while the worst-performing plots (ARI < 0.45) featured dense, multi-layered canopies with 38–62 overlapping trees (Fig. 7 and 8). Notably, the worst-performing plot had 62 ground truth trees but only 34 predicted trees—a 45% under-segmentation rate—despite being one of the most species-rich plots (21 identified species). This suggests that species diversity and structural complexity, rather than within-plot tree density, challenge the seed detection algorithm.

The density-based seed detection, while effective for well-separated trees, is fundamentally limited by the spatial resolution of the density grid (*density_cell*, Table 2). In dense stands, the local density peaks from adjacent trunks merge, creating a single peak that corresponds to multiple trees. This is exacerbated by variable trunk diameters and irregular spacing. This challenge is common to all bottom-up segmentation methods that rely on detecting individual stems from point clouds, including approaches such as *3D Forest* (Trochta et al., 2017) and *treeseg* (Burt et al., 2019), which also face difficulties in dense, overlapping stands. Future work should explore alternative seed detection strategies, including hierarchical clustering of minima, adaptive density thresholds, and integration of allometric constraints.

6.4. Robustness across forest types and point densities

Despite the seed detection limitations in dense stands, *TopTreeSegR* demonstrated remarkable robustness across diverse forest types and point densities. The mean ARI of 0.695 ± 0.118 across 30 plots from six Polish forests—ranging from open coniferous stands (Herby) to dense deciduous forests (Katrynka) and mixed species plots (Gorlice, Suprasl)—indicates that the topological framework generalizes well across ecological conditions. Notably, the strong tree count agreement (R^2 = 0.964; Fig. 6d) suggests that even when boundary assignment is imperfect, *TopTreeSegR* accurately detects the number of trees present—a critical requirement for operational inventory.

Processing time showed considerable variability (331–7,449 seconds), with only a weak correlation with point count (r = 0.421), indicating that structural complexity—rather than data volume—is the primary driver of computational cost. The outlier plot (7,449 seconds) exhibited the longest processing time, driven primarily by its unusually long preprocessing phase (5,657 seconds), which accounted for 76% of the total processing time. This suggests that preprocessing, particularly ground classification and normalization, may become computationally expensive in dense, multi-layered canopies where ground visibility is limited. Optimizing these preprocessing steps could substantially reduce processing time for complex plots.

Data reduction from preprocessing averaged 95.1% across all plots, reflecting both decimation and the removal of non-tree points. This step currently relies on ground truth labels (treeID != 0). The development of an automatic tree/non-tree classifier would be a significant advance, enabling *TopTreeSegR* to process raw TLS point clouds without requiring pre-filtered tree points, dramatically expanding its operational utility.

6.5. Comparison with existing methods

Direct quantitative comparison with other open-source methods is not possible, as they operate on point clouds rather than mesh representations. However, we can contextualize our results relative to published performance in similar forest types.

The mean ARI of 0.695 achieved by *TopTreeSegR* across 30 diverse plots is comparable to or exceeds reported performance of open-source methods in similar temperate forests. Trochta et al. (2017) reported a mean F1 score of approximately 0.60–0.75 for *3D Forest* across their benchmark datasets, while Burt et al. (2019) reported tree detection rates of 70–96% for *treeseg* depending on forest complexity. Our mean F1 of 0.788 and ARI of 0.695 across a wider range of forest types suggests competitive performance.

TopTreeSegR offers two key advantages over these methods. First, parameter-free operation: unlike *3D Forest* and *treeseg*, which requires extensive parameter tuning for each forest type, *TopTreeSegR* operates with fixed default parameters across all 30 plots. Second, topological rigor: the discrete Morse theory framework provides a mathematically well-founded approach to handling overlapping canopies and complex branching structures, which point-based methods often address through heuristic rules.

Conversely, the seed detection limitation in dense forests identified in Section 6.3 affects all bottom-up methods, including *3D Forest* and *treeseg*, which also struggle with overlapping trunks. Deep learning-based approaches like *FSCT* (Krisanski et al., 2021) may offer advantages in extremely dense stands but require labeled training data

and substantial computational resources. Future work could explore hybrid approaches combining topological segmentation with deep learning-based seed detection.

6.6. Implications and limitations

The plot-wise optimization of *TopTreeSegR* has significant practical implications for operational forest inventory. Processing times compatible with field campaign schedules (mean = 1,213 seconds per plot), combined with zero-parameter tuning, suggest integration into routine inventory workflows is feasible. The strong tree count agreement ($R^2 = 0.964$) indicates reliable stem density estimation across diverse forest types. The R-based implementation lowers the barrier to adoption, enabling forest ecologists to process their own TLS data without extensive computational training. The open-source nature facilitates community contributions and continuous improvement.

Several limitations should be acknowledged. First, validation was conducted on 30 plots from Polish forests; performance on other forest types (tropical, boreal, Mediterranean) remains to be tested. Second, the computational demands of α-complex construction and discrete Morse theory scale super-linearly with point count for extremely large plots (>10 million points). While decimation mitigates this, future work could explore distributed implementations leveraging the Stellar tree data structure (Fellegara et al., 2021), which provides a compact, efficient representation for simplicial complexes, combined with GPU-accelerated discrete Morse theory computation. Additionally, adaptive decimation strategies and processing full-resolution point clouds could further enhance segmentation accuracy in complex forest structures. Third, evaluation metrics focus on mesh vertex-level accuracy and tree counts; future validation should extend to tree-level structural metrics (height, DBH, crown volume) between segmented meshes and inventory data.

7. Conclusions

We have presented *TopTreeSegR*, a mesh-based topological segmentation framework that addresses the critical bottleneck of automated individual tree extraction from plot-wise TLS point clouds. By constructing an α-complex mesh representation and applying discrete Morse theory, our approach provides a mathematically rigorous foundation for partitioning forest canopies into individual trees. The framework operates with zero parameter tuning across diverse forest types, making it accessible to forest ecologists and practitioners without specialized computational expertise.

Our findings demonstrate that *TopTreeSegR* achieves robust segmentation accuracy (mean ARI = 0.695) across 30 diverse Polish forest plots, with processing times (mean = 1,213 seconds per plot) compatible with operational inventory workflows. The critical limitation identified—the failure of density-based seed detection in dense, overlapping stands—points toward clear future research directions. This challenge affects all bottom-up segmentation methods. Addressing it will require novel approaches, including hierarchical clustering of minima, adaptive density thresholds, integration of allometric constraints, or hybrid strategies combining topological analysis with deep learning-based seed detection.

The mesh-based topological paradigm offers broader implications beyond segmentation. The α-complex representation captures not only tree structure but also the connectivity relationships of the entire forest canopy. This topological information could inform downstream applications such as species identification, biomass estimation, and ecological monitoring, as canopy structure is closely linked to these ecological attributes. The open-source R implementation provides a platform for the remote sensing community to build upon and extend this foundational work.

In conclusion, *TopTreeSegR* represents a significant advance in automated individual tree segmentation from TLS point clouds. By combining mesh-based topological analysis with plot-wise optimization and open-source accessibility, it addresses key barriers that have limited TLS adoption in operational forest monitoring. While challenges remain in extremely dense forests, the framework's demonstrated robustness, accuracy, and parameter-free operation position it as a valuable tool for the forest ecology and remote sensing communities. Continued development—particularly in automatic tree/non-tree classification and adaptive seed detection—will further enhance its utility and generality, enabling integration of TLS into national and global forest monitoring programs.

8. Data and code availability

The TreeScanPL10k dataset used in this study is available from the original publication (Stereńczak et al., 2026). The Wytham Woods reference dataset is available from Calders et al. (Calders et al., 2022).

The *TopTreeSegR* R package is freely available on GitHub: https://github.com/DijoG/TopTreeSegR. The underlying mesh construction library, *ahull3D*, is available on GitHub and PyPI (*ahull3Dpy*). A Python implementation, *TopTreeSegpy*, is under active development and will be released upon completion of the *DiscreteMorsepy* library. All scripts used for analysis and figure generation are available in the supplementary material.

9. Authorship contribution statement

Gergő Diószegi: Conceptualization, Data curation, Methodology, Software, Validation, Visualization, Writing – original draft. **Dávid Abriha**: Writing – review & editing. **Szilárd Szabó**: Writing – review & editing.

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